

Preliminary Project Report 2:

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Topic: AI-Assisted Forensic Enhancement and Minutiae Localization in Partial Latent Fingerprints Using Pix2Pix Networks

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Github : <https://github.com/sdShaggy/GAFIS>

Outcomes expected : Presentable Literature + prototype

Dataset Preparation and Analysis for Training Pix2Pix Networks

Introduction:

The effectiveness of any fingerprint enhancement framework is strongly dependent on the quality and diversity of the training dataset. Therefore, the initial phase of this project focused on dataset preparation, statistical analysis, image quality assessment, preprocessing, and synthetic degradation generation. The objective was to create a robust dataset suitable for subsequent deep learning-based fingerprint enhancement experiments.

Dataset Description:

The primary dataset utilized in this work is the SOCOFing (Sokoto Coventry Fingerprint Dataset) Real Fingerprint Database. The dataset contains fingerprint impressions collected from multiple individuals and provides high-quality ridge and valley structures suitable for enhancement studies.

For this study, the Real subset of SOCOFing was employed. The dataset consists of 600 subjects, with each subject contributing ten fingerprint impressions corresponding to different fingers.

The dataset was selected due to its balanced representation of fingerprint patterns, standardized acquisition conditions, and suitability for synthetic latent fingerprint generation.

Dataset Statistics:

A statistical analysis was performed to understand the characteristics of the dataset.

Subject Distribution

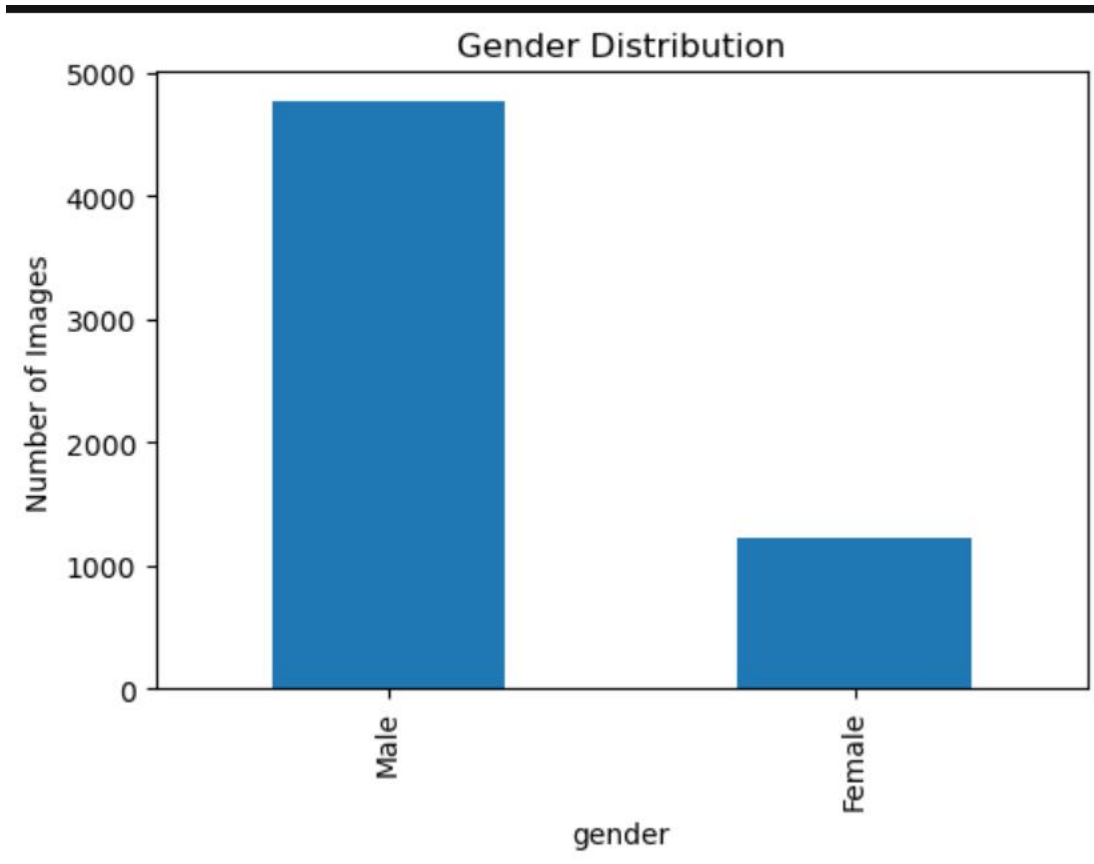
Parameter	Value
Total Subjects	600
Total Images	6000
Images per Subject	10

The dataset provides a uniform distribution of images across all subjects, minimizing subject-level sampling bias.

Gender Distribution

Gender	Number of Images
Male	4770
Female	1230

Male fingerprint samples constitute approximately 79.5% of the dataset, while female samples account for approximately 20.5%.



Resolution Analysis:

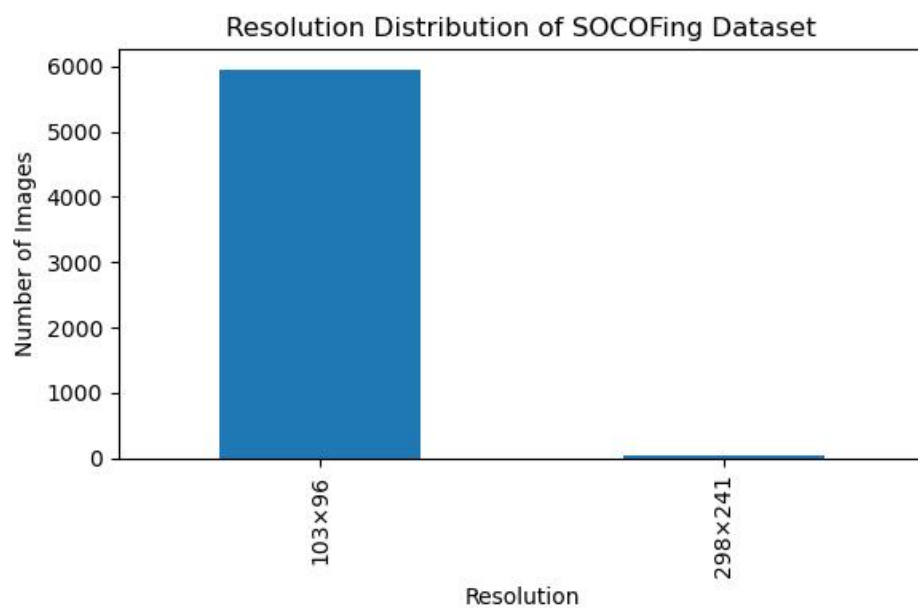
Image resolution analysis was conducted to determine dataset consistency.

Resolution	Number of Images
103 × 96	5956
298 × 241	44

Results indicate that approximately 99.27% of images possess a resolution of 103 × 96 pixels.

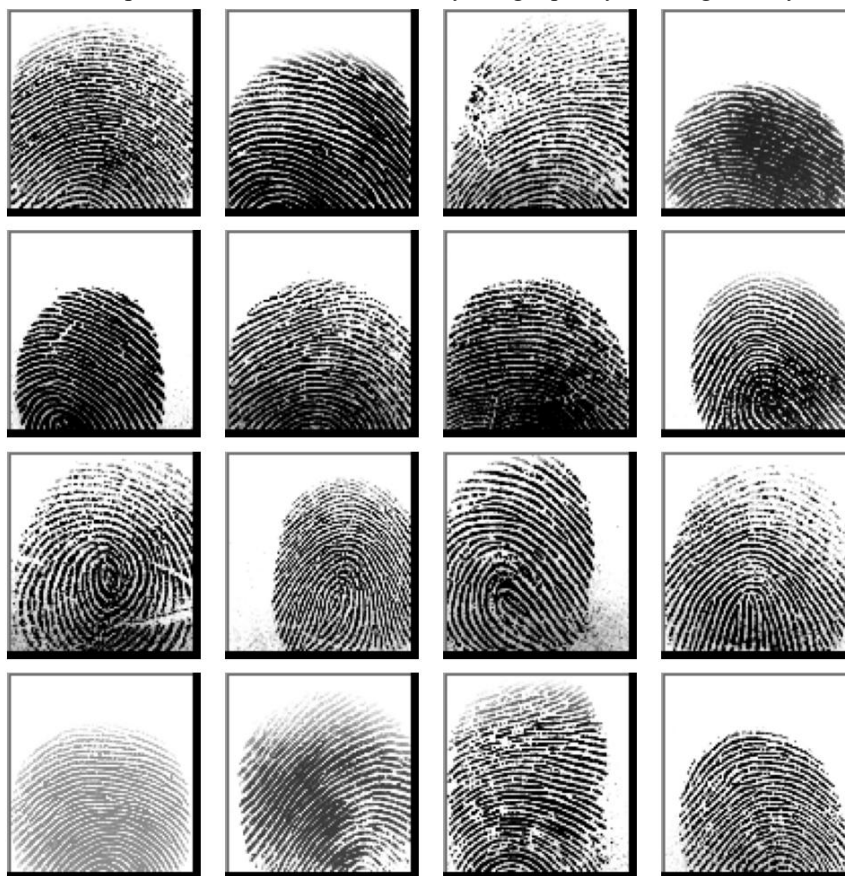
To ensure uniformity during preprocessing and future model training, all images were resized to 256 × 256 pixels.

Resolution analysis revealed that the dataset is highly consistent in terms of image dimensions. Approximately 99.27% of fingerprint images possess a resolution of 103 × 96 pixels, while only 0.73% of images exhibit a higher resolution of 298 × 241 pixels. Due to this minor variation, all images were resized to a uniform resolution of 256 × 256 pixels during preprocessing to ensure consistency across subsequent enhancement and learning stages.



Representative Dataset Samples:

A visual inspection was conducted to verify image quality and ridge clarity.



Motivation:

Fingerprint enhancement algorithms rely heavily on ridge clarity and image quality. Therefore, a quantitative quality assessment was performed before preprocessing.

The following metrics were utilized:

- Mean Intensity
- Contrast
- Entropy
- Blur Score (Variance of Laplacian)

Quality Metrics:

The statistical summary of the dataset is shown below.

Metric	Mean	Standard Deviation
Mean Intensity	142.78	22.96
Contrast	103.81	6.66
Entropy	3.61	0.40
Blur Score	58055.87	16871.98

Mean Intensity

The average intensity value indicates moderate brightness levels throughout the dataset, suggesting consistent image acquisition conditions.

Contrast

The observed contrast values demonstrate strong ridge-valley separation, making the dataset suitable for enhancement experiments.

Entropy

Entropy values indicate sufficient structural information within fingerprint ridge patterns.

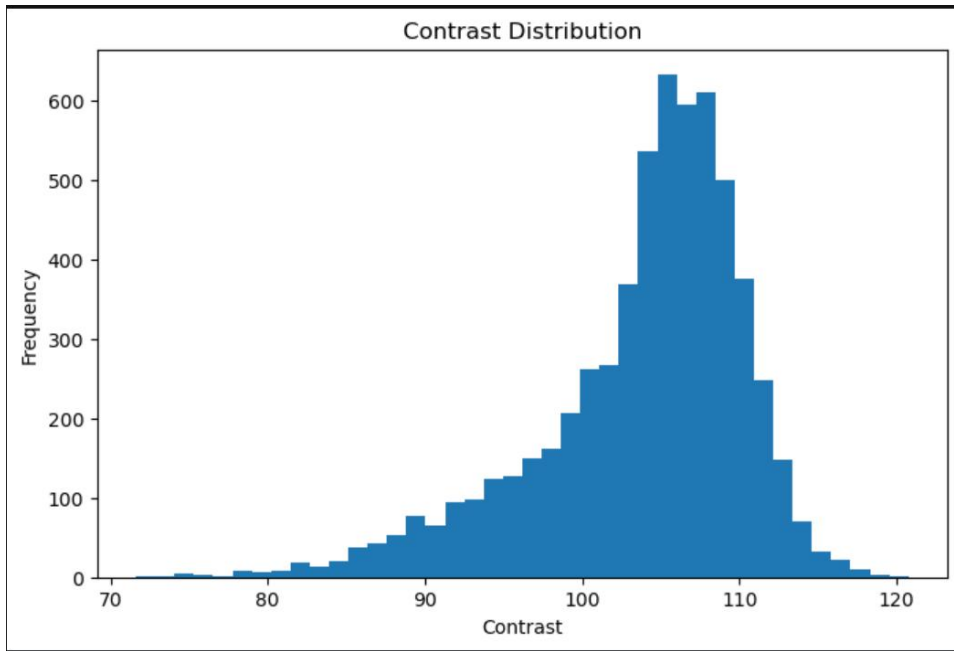
Blur Score

The high blur score values suggest that the majority of fingerprint images are sharp and contain well-defined ridge structures.

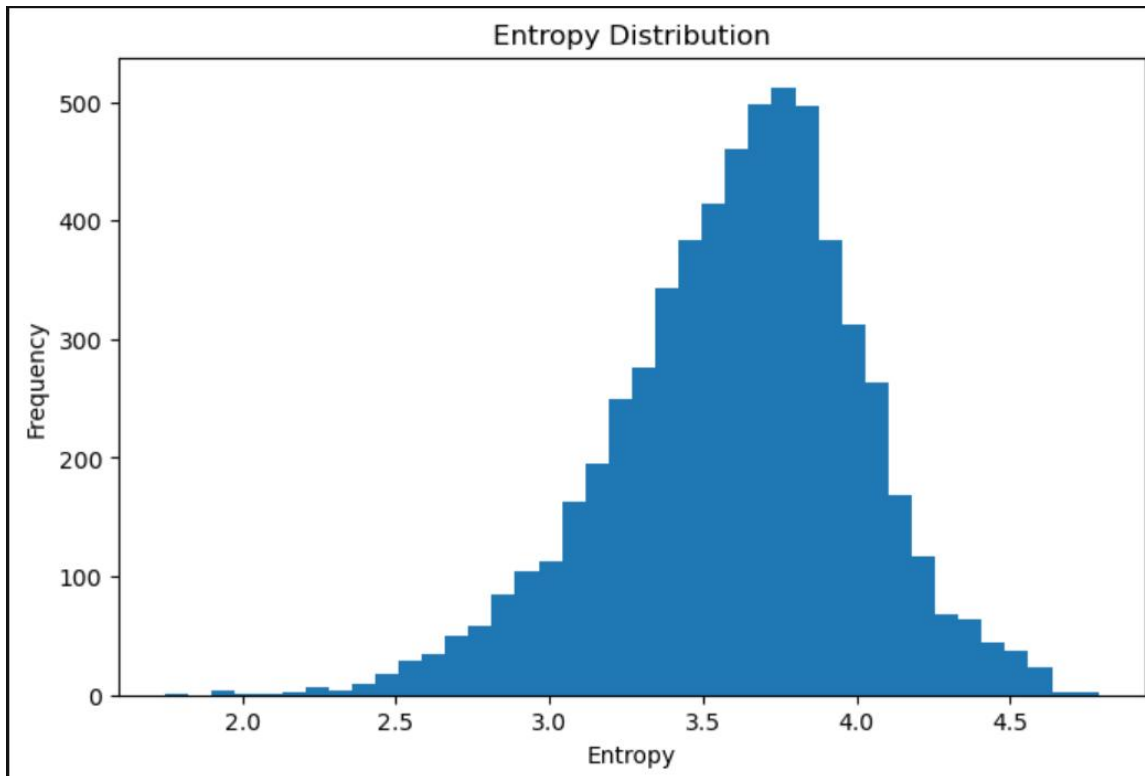
Quality Metric Distributions:

The distributions of quality metrics were analyzed to evaluate dataset consistency.

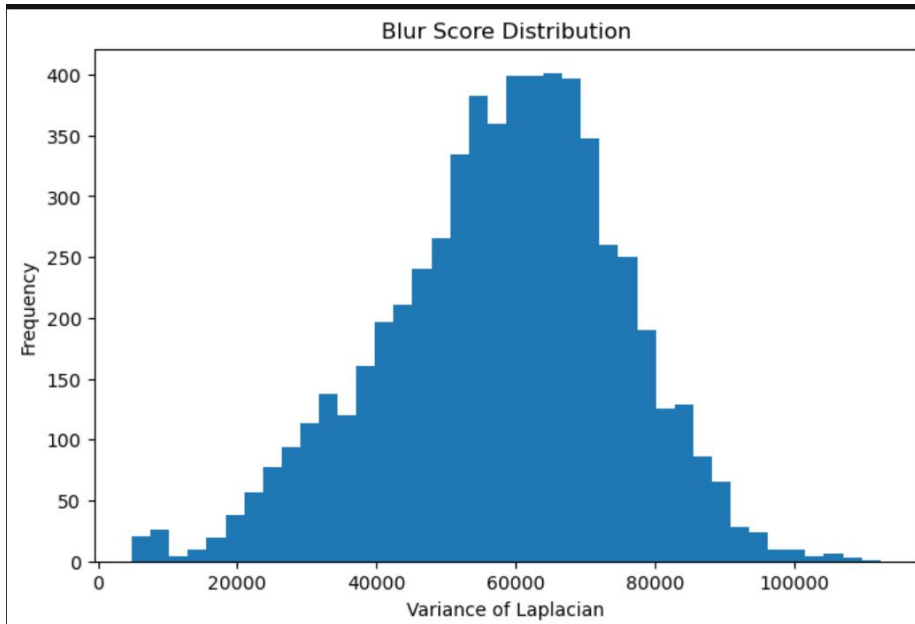
Contrast Distribution:



Entropy Distribution:



Blur Score Distribution:



Fingerprint Preprocessing:

Objective:

The preprocessing stage aims to improve fingerprint visibility and normalize image characteristics before enhancement and deep learning-based processing.

Preprocessing Pipeline:

The following preprocessing pipeline was developed:

Fingerprint Image

↓

Resize (256×256)

↓

CLAHE (Contrast Limited Adaptive Histogram Equalization)

↓

Intensity Normalization

↓

Edge-Based Ridge Visualization

CLAHE(Contrast Limited Adaptive Histogram Equalization) Enhancement:

Fingerprint images often suffer from non-uniform illumination and local contrast variations that can reduce the visibility of ridge and valley structures. To address this issue, Contrast Limited Adaptive Histogram Equalization (CLAHE) was applied during preprocessing. CLAHE improves local contrast

by dividing the image into small regions called tiles and performing histogram equalization independently within each region. Unlike conventional histogram equalization, CLAHE incorporates a contrast limiting mechanism that prevents excessive amplification of noise and artifacts.

In this study, CLAHE was implemented using a clip limit of 2.0 and a tile grid size of 8×8 . The method enhanced ridge visibility while preserving important fingerprint structures. Improved local contrast facilitates subsequent processing stages such as synthetic degradation generation, ridge enhancement, minutiae extraction, and deep learning-based fingerprint reconstruction.

Intensity Normalization:

Normalization was applied to standardize pixel intensity values across all fingerprint images.

This step reduces illumination variation and improves consistency for future model training.

Preprocessing Pipeline Demonstration

Original Print → *Resize(256x256)* → *CLAHE* → *Intensity Normalization* → *Edge-Based Ridge Visualization*



Synthetic Latent Fingerprint Generation:

Motivation:

Obtaining large-scale paired latent and rolled fingerprint datasets is challenging due to privacy restrictions and limited public availability.

To address this limitation, a synthetic degradation framework was developed to simulate latent fingerprint acquisition conditions.

Synthetic Degradation Framework:

The framework applies controlled degradations to clean fingerprint images.

The degradation pipeline includes:

- Gaussian Blur
- Contrast Reduction
- Partial Occlusion
- Information Loss Simulation

These transformations mimic common forensic latent fingerprint challenges such as smudging, incomplete impressions, and poor surface contact.

Degradation Levels:

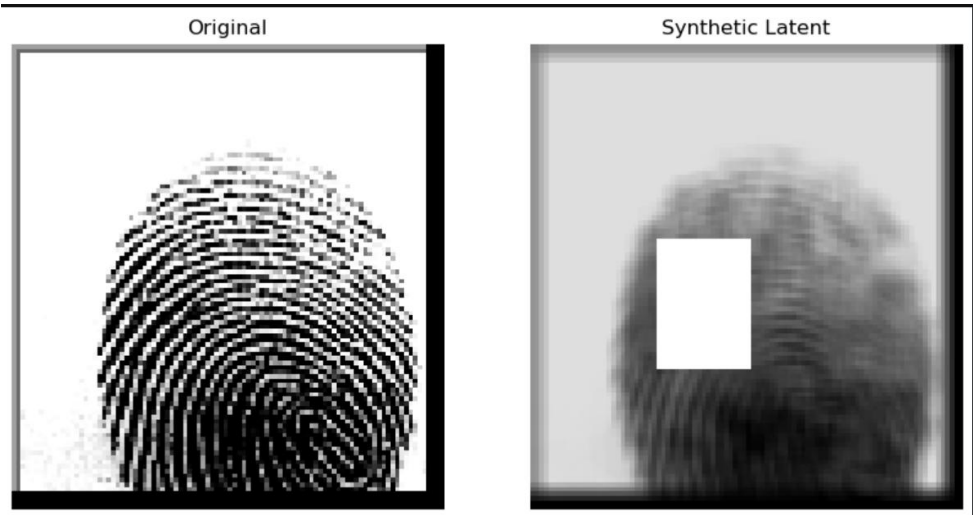
Three degradation levels were defined:

Level	Characteristics
Mild	Low blur, minimal occlusion
Moderate	Moderate blur and information loss
Severe	High blur and significant occlusion

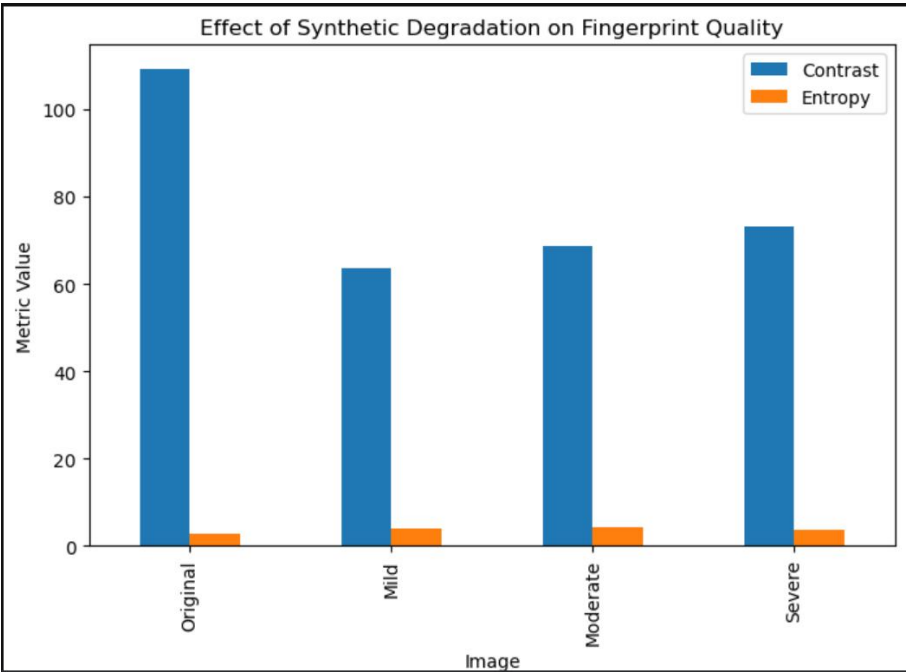
Experimental Results:

Visual examples of generated latent fingerprints are shown below.

Original vs Synthetic Latent Fingerprint



Mild, Moderate, and Severe Degradation Levels:



Quantitative Evaluation:

Quality metrics were computed for degraded fingerprints.

Results demonstrate a consistent reduction in contrast, entropy, and blur score as degradation severity increases.

This confirms that the proposed degradation framework successfully simulates the loss of fingerprint information encountered in forensic latent prints.

Quality Metric Comparison Across Degradation Levels:

```
metrics_df = pd.DataFrame(  
    results,  
    columns=columns  
)  
  
print(metrics_df)  
  
metrics_df
```

[40] ✓ 0.1s Python

	Image	MeanIntensity	Contrast	Entropy	BlurScore
0	Original	161.05	109.28	2.96	54712.86
1	Mild	113.18	63.65	4.10	382.90
2	Moderate	117.80	68.58	4.18	460.05
3	Severe	143.53	73.21	3.63	1333.41

Conclusion

This phase of the project established a complete fingerprint data preparation and preprocessing framework using the SOCOFing dataset.

A comprehensive statistical analysis was performed, image quality characteristics were evaluated, and a preprocessing pipeline was developed to enhance ridge visibility. Furthermore, a synthetic degradation framework was implemented to generate latent-style fingerprints for future supervised enhancement experiments.

The resulting dataset and preprocessing methodology provide a strong foundation for the next phase of the project, which will focus on deep learning-based fingerprint enhancement and forensic matching.

The next phase of this work will include the training of GAN Pix2Pix based on the synthetically degraded SOCOFing fingerprint images and will comprise of Kaggle Fingerprint Dataset based Validation.

The Results in this report can be verified at :

https://github.com/sdShaggy/GAFIS/blob/main/services/SOCOFing_processing.ipynb